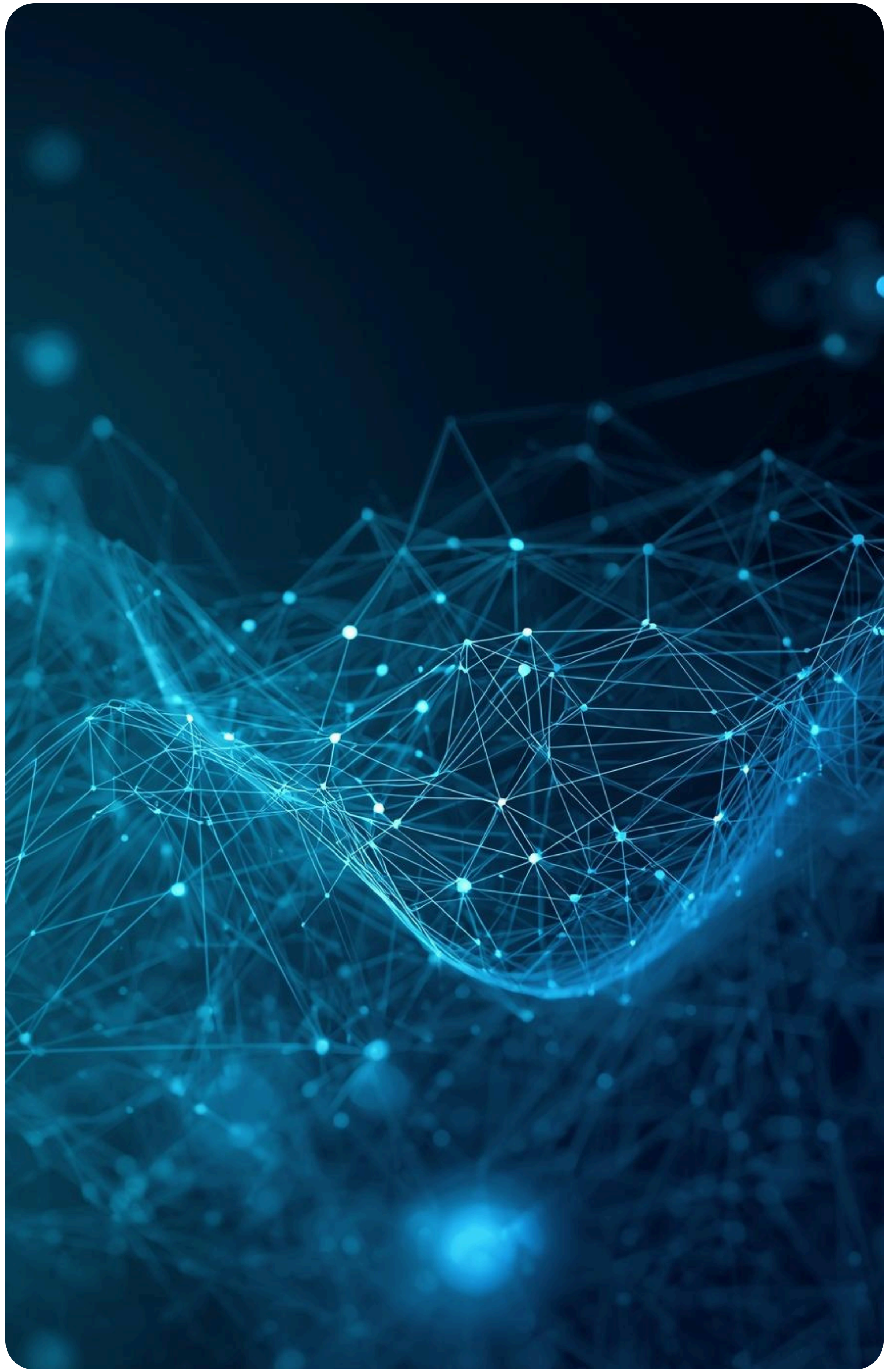
Safety Research Directions



Adversarial Robustness

- Training on adversarial examples to increase resilience
- Certified defenses with mathematical guarantees
- Detection of adversarial inputs
- Challenge: Arms race between attacks and defenses

Interpretability and Transparency

- Understanding what neural networks learn and why they make decisions
- Mechanistic interpretability: reverse-engineering model internals
- Feature visualization, attribution methods, concept-based explanations
- Goal: Enable debugging, verification, and trust

Formal Verification

- Mathematical proofs that systems satisfy safety properties
- Works well for simple systems, scales poorly to complex neural networks
- Hybrid approaches combining formal methods with ML

Safe Exploration and Testing

- How to test AI systems without real-world harm
- Simulation environments, red-teaming, controlled deployment
- Challenge: Simulators imperfect, testing can't cover everything

Human-AI Interaction Safety

- Designing systems that fail gracefully
- Clear communication of limitations and uncertainties
- Monitoring and intervention mechanisms
- Challenge: Automation bias, complacency, responsibility gaps